Sortify Application Test Cases

Seven executed feature-specific test cases covering waste recognition, confidence fallback, disposal guidance, assistant responses, history, feedback, and admin catalog control.

Project name	Sortify	Class / roll no.	UCS503P	App version	Sortify prototype	Environment	Prototype application	Execution date	31-Oct-2026

Test case ID	Module	Scenario / objective	Test type	Preconditions	Test data	Test steps	Expected result	Actual result	Status	Priority	Evidence / defect ID
SORT-SCAN-001	Waste Recognition	Identify a clearly visible recyclable item and display the mapped disposal result	Positive	Sortify is open. The detector service is reachable and the ten-class model is loaded. Camera or image upload permission is available.	Image: clean empty plastic water bottle; expected model class: plastic_bottle; expected disposal stream: Recyclable.	1. Open the Scan screen and choose Photo. 2. Capture or upload the test image. 3. Start identification and wait for inference to finish. 4. Inspect the result label, category, confidence and bounding box.	The result identifies the item as a plastic bottle and maps it to the Recyclable stream. Confidence is displayed, the detector box is positioned over the detected item, and relevant preparation/disposal guidance is shown. No unrelated category is displayed.	The plastic bottle was correctly identified and mapped to the Recyclable stream. Confidence and the detection bounding box were displayed, along with the corresponding preparation and disposal guidance. No unrelated category was shown.	**Pass**	High	None
SORT-SCAN-002	Recognition Fallback	Handle a low-confidence or ambiguous image without presenting an unreliable result as certain	Negative	Sortify is open. Samples mode and its Demo options are available, including the Low-confidence result switch.	Demo sample: Low-confidence result enabled; ambiguous item image/sample.	1. Open Scan and switch to Samples. 2. Open Demo options and enable Low-confidence result. 3. Run the sample scan. 4. Review the result and available recovery actions. 5. Select a suggested alternative or use manual search, where offered.	The app communicates that the prediction is uncertain and presents ranked alternatives or a manual-search fallback. It must not silently treat the low-confidence prediction as a confirmed classification or provide misleading definitive disposal instructions.	The low-confidence sample displayed an uncertainty state and offered alternative suggestions/manual search rather than presenting the prediction as certain. The user could proceed through the available recovery path without receiving definitive, unsupported disposal advice.	**Pass**	High	None
SORT-GUIDE-001	Disposal Guidance	Show preparation steps and safety guidance associated with a hazardous waste classification	Positive	A hazardous item exists in the catalog and its disposal guidance is configured. The scan result screen is accessible.	Item: household battery; mapped class: battery_ewaste; expected stream: E-Waste; guidance: keep separate from regular waste and use an appropriate e-waste collection route.	1. Scan or select a sample image of a battery. 2. Open the generated result. 3. Inspect the waste stream, preparation steps, safety notes and area-specific context. 4. Navigate away and return to the result if the screen supports it.	The item is mapped to the E-Waste stream. Relevant safety notes and preparation/disposal guidance are visible on the result screen, rather than generic household-waste advice. The displayed guidance corresponds to the selected catalog item.	The battery was mapped to the E-Waste stream. The result displayed the relevant preparation instructions and safety guidance, including keeping it separate from regular waste and using an appropriate e-waste collection route.	**Pass**	High	None
SORT-ASSIST-001	Sortify Assistant	Answer a disposal question using the supported waste guide and provide a useful fallback for an unknown item	Boundary	The Assistant screen is available and the local waste guide contains entries for supported items. Conversation history is enabled in local storage.	Query 1: "Can I recycle a greasy pizza box?" Query 2: "How do I dispose of an imaginary alloy foam?"	1. Open Sortify Assistant. 2. Enter Query 1 and send it. 3. Verify that the reply references relevant guide information. 4. Send Query 2 in the same conversation. 5. Inspect the fallback response and conversation thread.	For the supported/recognised query, the assistant returns a relevant answer grounded in catalog guidance and links or points to the relevant guide entry. For the unknown query, it clearly indicates that it cannot determine a reliable instruction and suggests browsing/searching the guide instead of inventing disposal advice. Both messages remain in the conversation.	The assistant returned relevant guidance for the greasy pizza-box question based on the supported guide. For the unknown-item query, it indicated that it could not provide a reliable instruction and directed the user to search/browse the guide. Both exchanges remained visible in the conversation.	**Pass**	High	None
SORT-HISTORY-001	Scan History	Persist a completed scan and its sorting state across app reloads	Positive	Sortify is open with local storage available. A sample or live scan can be completed and guest mode is permitted to save history.	Test item: cardboard box; classification: cardboard; action: save item, then mark as sorted.	1. Complete a scan for the cardboard sample. 2. Save the result when prompted. 3. Mark the saved item as sorted. 4. Open History and verify the entry and status. 5. Reload or restart the app and revisit History.	The saved scan appears in History with the correct item/category and its sorted state. The record remains after reload/restart and contributes to the relevant local activity counts. The app does not create duplicate entries merely because History is reopened.	The cardboard scan was saved to History and displayed with the correct item/category and sorted status. The entry remained available after reloading the app, and reopening History did not create a duplicate record.	**Pass**	Medium	None
SORT-FEEDBACK-001	User Feedback	Submit a correction for a misclassified waste item and make it available for review	Positive	A scan result is displayed and the correction/feedback control is available. The app has local storage enabled.	Original prediction: plastic_bottle. User correction: metal can. Feedback reason: Incorrect item identification.	1. Open a scan result whose predicted item is incorrect for the test. 2. Select the correction/feedback action. 3. Choose or enter the corrected item/category and submit. 4. Open the relevant feedback review area (admin) or inspect the saved feedback state. 5. Reload the app and verify the correction remains recorded.	The correction is accepted and stored as feedback associated with the scan. It is visible in the admin feedback review workflow and retains the original prediction plus the submitted correction. Submitting feedback does not silently rewrite the classifier output or mark the item sorted.	The correction was submitted successfully and appeared in the admin feedback review area with the original prediction and user-provided correction. The feedback persisted after reload and did not alter the classifier's original prediction or mark the item as sorted.	**Pass**	Medium	None
SORT-ADMIN-001	Admin Catalog Management	Update a catalog entry and verify that the revised disposal guidance is used by the app	Positive	The prototype Admin screen is accessible. A catalog item can be edited and changes are persisted in local storage.	Existing catalog item: plastic bottle. Change: update preparation guidance to state that the container should be emptied/rinsed and the cap handled according to local rules.	1. Open the Admin screen and locate the plastic bottle catalog entry. 2. Edit its preparation/disposal guidance and save the change. 3. Return to the Waste Guide and search for plastic bottle. 4. Open the matching item and verify the revised guidance. 5. Reload the app and verify the edit persists.	The catalog edit is saved and the Waste Guide displays the updated guidance for the same item after navigation and reload. The change is reflected without modifying or retraining the image-classification model.	The plastic-bottle catalog entry was updated successfully. The Waste Guide displayed the revised preparation guidance, and the change remained after reloading the app. The catalog edit did not require changing or retraining the recognition model.	**Pass**	Medium	None